O–H–204a
February 2, 1966
SUPERSEDING
Fed. Spec. O-H-204
July 15, 1965

FEDERAL SPECIFICATION

HERBICIDE: DISODIUM METHANEARSONATE, DMA

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the water soluble herbicide, 63 percent disodium methanearsonate (DMA), equivalent to 100 percent DMA hexahydrate. It is used diluted with water as a selective herbicide primarily for post emergence control of crabgrass. It can also be used in controlling the growth of Dallis grass in Bermuda grass and Zoysia grass.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

TT-I-558—Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metals, Glass, etc.).
PPP-B-35—Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-A-3941—Adhesive, Paper Label, Water-Resistant.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific

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procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations:

7 CFR 362—Agriculture, Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act.

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Chemical properties. When tested as specified in 4.5, the material shall conform to table I.

TABLE I.

Property	Requirement, percent by weight		Applicable test paragraph
	Min.	Max.	
Disodium methanearsonate	63.0	—	4.5.1
Inert material	—	37.0	4.5.2
Total arsenic (As As)	25.5	—	4.5.3

3.2 pH. The herbicide shall exhibit a pH of no less than 10.0 nor more than 11.5 when tested as specified in 4.5.4.

3.3 Appearance. The herbicide shall be a free flowing powder which shall be dry, loose, free from lumps and foreign matter, and pour smoothly in an unimpeded stream when tested as specified in 4.5.5.

3.4 Workmanship. The finished product shall be free from any defect which might impair its utility. It shall conform to all requirements specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 *Herbicide.* For purposes of inspection, a lot shall consist of all herbicides produced by one manufacturer under essentially the same conditions and process, within the following limitations. In the event the herbicide is produced by a continuous-run process, the lot shall contain herbicide produced in a period of not more than 24 hours. In the event the herbicide is produced by a batch process, each batch shall constitute a lot. Herbicide in the inspection lot shall be identified by order of production when a continous-run process is involved and by a batch number when a batch process is involved.

4.2.2 *Filled containers.* For purposes of inspection, a lot shall consist of all specified containers filled with approved herbicide as required and which are to be offered for delivery at one time under a contract or order.

4.3 Sampling.

4.3.1 *Sampling for tests.* Three separate one-pound samples of the herbicide shall be taken from each inspection lot. When herbicide is produced by a continuous-run proc-

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ess, the three samples shall be taken so as to represent, respectively, the first part, the middle part, and the last part of the run which constituted the inspection lot. When herbicide is produced by a batch process, the three samples shall be taken at different locations within each batch.

4.3.2 *Sampling for examination of filled containers.* A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, at inspection level I, and an acceptable quality level (AQL) of 2.5 percent defective.

4.4 **Examination.**

4.4.1 *Filled containers.* Each sample filled container selected in accordance with 4.3.2, shall be examined for any defects of the container and the container closure, for evidence of leakage, and for illegible, incorrect, or missing marking. Each sample filled container shall also be weighed to determine the net weight of contents. Any filled container having one or more defects shall be considered a defective unit.

4.5 **Tests.** Samples of herbicide selected in accordance with 4.3.1 shall be subjected separately to the tests specified herein. If any one of the samples fails to pass any one of the tests specified, the lot represented therein shall be rejected. Rejected lots may be reinspected provided the manufacturer has corrected all defects. Prior to resubmission of rejected lots, the manufacturer shall furnish full particulars to the Government concerning the rejection and the action taken to correct the defects.

4.5.1 *Determination of disodium methanearsonate.* Disodium methanearsonate has two neutralization points when titrated against standard acid. The first neutralization point (approximately pH 6.4) corresponds to a dilute solution of sodium acid methanearsonate, the second neutralization point (approximately pH 2.6) corresponds to a dilute solution of methanearsonic acid. The percent DMA present is calculated from the volume of acid used in reaching the second neutralization point since traces of As_2O_3 or As_2O_5 are titrated as their sodium salts in the first titer.

4.5.1.1 *Procedure.* Dissolve 2.5 grams (g.) herbicide in 100 milliters (ml.) distilled water contained in a 250 ml. Erlenmeyer flask. Using a recording titrimeter or a pH meter, add 1.0 *N* HCl to the flask until the first inflection point previously determined is reached at pH of approximately 6.4. Record amount 1.0 *N* HCl used. Add 1.0 *N* HCl until the second inflection point previously determined has been reached. Using the HCl titer obtained between first and second inflection points, calculate the percent DMA present as follows:

$$\text{Percent } Na_2CH_3AsO_3 \cdot 6H_2O = \frac{(\text{ml. HCl})\ (N\ \text{HCl})\ (0.1839) \times 100}{\text{g. sample}}$$

4.5.2 *Inert material.* Inert material will be reported as the difference between 100 percent and the percent DMA found.

4.5.3 *Determination of total arsenic.* Dissolve 0.2 g. of herbicide with 10 ml. HNO_3 (1 + 4) in an evaporating dish. Add 5 ml. concentrated H_2SO_4 and heat on a hotplate until dense white fumes of SO_3 just appear in the flask. Cool, add 10 ml. distilled water and again evaporate until SO_3 fumes appear. Complete removal of HNO_3 is indicated by clarity of the solution and absence of brownish fumes. Carefully transfer into a 250 ml. Erlenmeyer flask. Let cool and dilute to approximately 100 ml. final volume. Add 1 g. KI and boil until volume is reduced to 40 ml. Cool, dilute to 100 ml. Add 0.05 *N* sodium thiosulphate dropwise until iodine color is removed. Neutralize with 6 *N* NaOH. Add 4 g. $NaHCO_3$ and titrate with 0.1 *N* iodine until yellow color disappears. Add 5 ml. starch indicator and continue titration to a permanent blue color.

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4.5.3.1 *Calculation.*

$$\text{As, percent} = \frac{(\text{ml. } I_2)\ (\text{N of } I_2)\ (0.03746)\ (100)}{\text{Wt. of sample}}$$

4.5.4 *Determination of pH value.* 2.5 g. of herbicide shall be dissolved in 100 ml. CO_2-free distilled water and the pH determined at 25°C., using a pH meter.

4.5.5 *Appearance.* Measure 50 ml. of the dry herbicide into a glass cylinder seven inches high by 7/8 inch in diameter and equipped with a spout. By tilting the cylinder pour the herbicide at a uniform rate of approximately 100 ml. per minute. Observe the appearance of the herbicide in the cylinder and the pouring for conformance to 3.3.

4.6 **Inspection of preparation for delivery.** The packing and marking of the herbicide shall be inspected to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packing shall be in accordance with Fed. Std. No. 102.)

5.1 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* The herbicide shall be packed in 50 or 100 pound quantities in fiber drums conforming to PPP-D-723, type II, grade A, or textile laminated bags conforming to PPP-B-35, number P35B, as specified (see 6.2).

5.1.2 *Level B.* The herbicide furnished in 50 or 100 pound quantities as specified (see 6.2), shall be packed in fiber drums conforming to PPP-D-723, type I.

5.1.3 *Level C.* The herbicide shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with rules and regulations of carriers as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act of 1947, as amended, and in compliance with its requirements. The label data shall be durably and legibly marked in accordance with MIL-STD-129 or Fed. Std. No. 102, as applicable, on the main panel of the body by lithographing, stenciling or printing with a yellow, noncorrosive, weather-resistant, waterproof ink conforming to TT-I-558. Labels shall be securely affixed in place, with water-resistant label adhesive as specified in MIL-A-3941 and shall be waterproofed by coating the entire surface of the label with the same adhesive.

5.2.2 *Directions for use.* The following directions for use shall be included on each herbicide container:

DIRECTIONS FOR USE

Spray as soon as seedlings of crabgrass or Dallis grass are one to two inches high. Two or more applications are usually required at 7-10 day intervals between sprayings. If older plants are not killed or if new crabgrass or Dallis grass appears, repeat with one to two more additional sprayings. For best results treat early, apply accurately and do not apply during hot, dry spells unless grass is thoroughly watered. Grass roots should be watered daily after spraying, but do not soak. If moist from rain, grass need not be watered. DO NOT APPLY TO DRY TURF DURING HOT DRY SPELLS. RESTORE SOIL MOISTURE BY WATERING TURF WELL FOR A PERIOD OF TIME PRIOR TO SPRAYING.

YOUNG SEEDLINGS: Mix one-half pound with 20 gallons of water and apply to 4000 square feet.

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MATURE CRABGRASS OR DALLIS GRASS: Mix one-half pound with 20 gallons of water and apply to 2000-3000 square feet.

CAUTION

Harmful if swallowed.
Avoid contact with skin.
Avoid breathing dust.
Avoid storage near food or feed products.
Keep children and animals off treated areas.
Wash hands after using.

DO NOT APPLY TO BERMUDA GRASS INTENDED TO BE FED TO LIVESTOCK

5.2.3 *Civil agencies.* In addition to labeling specified in 5.2.1 and any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.4 *Military requirements.* In addition to labeling specified in 5.2.1 and any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Disodium methanearsonate is intended for use as a selective crabgrass killer at recommended rates of application in the presence of Kentucky, Merion Blue, Bermudas and Zoysia grasses. It will also control Dallis grass in the presence of Bermuda and Zoysia grasses and several undesired grasses (Johnson, water, goose and nut grasses), cocklebur and annual morning glory in the presence of cotton. Turf areas to be treated with this product include lawns, athletic fields, golf courses, and parks.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Inspection responsibility, methods and tests (section 4).
(*c*) Levels of packing required (5.1).

6.3 The Federal Insecticide, Fungicide and Rodenticide Act of 1947 is under the jurisdiction of the Pesticides Regulation Section, Plant Pest Control Branch, Agricultural Research Service, U. S. Department of Agriculture, Washington 25, D. C., to which agency all matters pertaining to registration should be addressed.

CUSTODIANS:

Army—MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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